

From Ordinary Differential Equations to Petri Nets and Gillespie Simulation

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1 From the ODE Model to a Stochastic Petri Net

The cXR-tXA regulatory network was originally formulated as a system of ordinary differential equations (ODEs), in which each state variable represents the normalized concentration of a biological species. This deterministic framework accurately describes the average evolution of the system but neglects the intrinsic stochasticity arising from the discrete and random nature of molecular interactions.

To capture these fluctuations, we construct a Stochastic Petri Net (SPN) whose dynamics constitute a stochastic counterpart of the original ODE model. The resulting model preserves the biological mechanisms encoded in the differential equations while enabling stochastic simulations through Gillespie’s Stochastic Simulation Algorithm (SSA).

The conversion from the continuous ODE formulation to the SPN proceeds in three successive steps:

1. discretizing each continuous state variable into an integer-valued molecular population;
2. representing every biological species by one or more places;

3. translating each production and degradation mechanism into an independent Petri net transition.

1.1 The Original ODE System

The original cXR–tXA model consists of six coupled ordinary differential equations describing the dynamics of the two *Xist* alleles, the two cis-acting repressors (cXR), and the two trans-acting activators (tXA).

Each equation can be written in the generic form

$$\frac{dx_i}{dt} = P_i(x) - D_i(x), \quad (1)$$

where $P_i(x)$ and $D_i(x)$ denote the production and degradation rates of species x_i , respectively.

Production rates are nonlinear functions involving Hill-type activation and inhibition terms, whereas degradation follows first-order kinetics and is proportional to the current concentration of the corresponding species.

This additive decomposition naturally separates the underlying biological mechanisms into production and degradation events, providing the basis for the Petri net representation developed in the following sections.

1.2 Discrete Representation of Biological Species

In the deterministic model, each biological species is represented by a normalized concentration belonging to the interval $[0, 1]$. In contrast, stochastic Petri nets operate on discrete tokens, each representing an individual molecule.

For every species i , we therefore introduce an integer-valued molecular population

$$M_i \in \{0, \dots, N_i\},$$

where N_i denotes the chosen maximum number of molecules for that species.

The correspondence between the continuous and discrete representations is given by

$$M_i = \lfloor N_i x_i \rfloor,$$

with x_i the normalized concentration of the ODE model.

Each biological species is represented by a dedicated place whose marking equals its current molecular population.

To facilitate the representation of inhibitory mechanisms and threshold conditions, complementary places are introduced whenever required. For a species S , the corresponding complementary place $\bar{S}$ satisfies the conservation relation

$$M(S) + M(\bar{S}) = N_S,$$

ensuring that the total number of tokens associated with the species remains constant throughout the simulation.

1.3 From Differential Equations to Petri Net Transitions

Once every state variable has been mapped to a place, the dynamical terms of the ODE model are translated into Petri net transitions.

Rather than associating one transition with each differential equation, every elementary production and degradation mechanism is modeled independently. Consequently, an equation of the form

$$\frac{dx}{dt} = P(x) - D(x)$$

is represented by two distinct transitions:

- a production transition, which increases the molecular population of the species;
- a degradation transition, which decreases it.

This decomposition reflects the stochastic interpretation of biochemical reactions, where production and degradation correspond to independent random events occurring with different firing rates.

For example, the dynamics of $Xist_1$ are governed by

$$\frac{dx_1}{dt} = P_{Xist_1}(x) - \delta x_1.$$

This equation gives rise to two transitions:

- t_1 , producing one molecule of $Xist_1$;
- t_2 , degrading one molecule of $Xist_1$.

The production transition increases the marking of the place associated with $Xist_1$, whereas the degradation transition decreases it.

Applying the same construction to the six differential equations yields a Petri net containing twelve transitions: one production and one degradation transition for each biological species.

1.4 Construction of the Pre- and Post-Incidence Matrices

After defining the transitions, the pre- and post-incidence matrices are constructed to specify both the enabling conditions and the state updates associated with each firing.

Their construction follows two complementary principles.

First, every species appearing in the regulatory functions of a production or degradation term is connected to the corresponding transition through a self-loop. Such places appear with identical weights in both the pre- and post-incidence matrices. Consequently, their tokens are temporarily consumed and immediately restored when the transition fires, leaving their marking unchanged while allowing the transition propensity to depend on their current molecular population.

Second, the species whose abundance is modified by the reaction undergoes an actual state change. Production transitions transfer one token from the complementary place to the species place, whereas degradation transitions perform the opposite operation. This construction preserves the total number of tokens associated with each species.

As an illustration, consider the production term of the first differential equation,

$$\frac{dx_1}{dt} = p_{21} f(cXR_1) f(T).$$

The function $f(cXR_1)$ is a decreasing Hill function describing the inhibitory effect of the cis-repressor. Instead of connecting the transition directly to the place representing cXR_1 , the complementary place $\overline{cXR_1}$ is used. A large marking of $\overline{cXR_1}$ corresponds to a low concentration of cXR_1 , precisely the biological condition favoring the production of $Xist_1$.

The activating factor T is represented by the two allelic species tXA_1 and tXA_2 . Both places are therefore connected to the transition through self-loops so that their molecular populations influence the firing propensity without being modified by the reaction.

Finally, the production of one $Xist_1$ molecule is represented by moving one token from the complementary place to the species place,

$$\text{Pre}(\overline{Xist_1}, t_1) = 1, \quad \text{Post}(Xist_1, t_1) = 1.$$

Consequently,

$$M(Xist_1) + M(\overline{Xist_1}) = N_{Xist_1}$$

remains invariant throughout the simulation.

The same construction principle is applied systematically to every production and degradation mechanism of the original ODE system. Regulatory species influence transition propensities through self-loops, whereas only the species directly involved in the reaction experience a change in their marking.

2 Stochastic Semantics of the Petri Net

Once the structure of the Petri net has been defined, the next step is to specify its stochastic semantics. In a Stochastic Petri Net (SPN), each transition that is enabled at a given marking is associated with a firing rate. This rate determines the probability that the transition fires in the next infinitesimal time interval.

In the proposed model, the stochastic behavior of each transition is determined by two complementary components:

1. an *intrinsic propensity function*, derived from the corresponding production or degradation term of the ODE model;
2. a *guard function*, encoding the threshold-based biological conditions under which the transition is admissible.

Together, these two components define the effective firing rate used in Gillespie's Stochastic Simulation Algorithm.

2.1 Transition Propensities

Each transition of the Petri net represents either a production or a degradation mechanism appearing in the original system of differential equations. It is therefore natural to define its propensity from the corresponding term of the ODE model.

For example, the dynamics of X_{ist_1} can be written as

$$\frac{dx_1}{dt} = P_{X_{ist_1}}(x) - D_{X_{ist_1}}(x).$$

This equation is decomposed into two stochastic events: one production event and one degradation event. Their intrinsic propensities are defined by

$$\lambda_{\text{prod}}(M) = P_{X_{ist_1}}(M),$$

and

$$\lambda_{\text{deg}}(M) = D_{X_{ist_1}}(M),$$

where the continuous concentrations of the ODE model are evaluated through the current marking M of the Petri net.

This choice is not arbitrary. The terms of the ODE already represent instantaneous production and degradation rates. The SPN interpretation replaces a continuous variation by a sequence of discrete molecular events whose average behavior matches the deterministic rate.

For instance, suppose that

$$\frac{dx}{dt} = 5.$$

In the deterministic model, this means that the quantity x increases at an average rate of five units per unit time. In the stochastic model, the same evolution is represented by individual production events occurring randomly, with an average frequency of five events per unit time.

This is achieved by assigning the production transition the propensity

$$\lambda(M) = 5.$$

If the waiting time before the next firing is exponentially distributed,

$$T \sim \text{Exp}(\lambda(M)),$$

then

$$\mathbb{E}[T] = \frac{1}{\lambda(M)}.$$

Thus, for $\lambda(M) = 5$,

$$\mathbb{E}[T] = 0.2,$$

which corresponds, on average, to five firings per unit time.

More generally, if each firing changes the marking by one molecule, then over a sufficiently small time interval dt ,

$$\mathbb{E}[\Delta M] = \lambda(M) dt.$$

Hence,

$$\frac{d\mathbb{E}[M]}{dt} = \lambda(M).$$

Therefore, choosing the propensity of each transition equal to the corresponding production or degradation term preserves the deterministic dynamics in expectation, while adding stochastic fluctuations around the mean trajectory.

2.2 Effective Transition Rates

In the simulation, the firing rate of a transition is not given by its intrinsic propensity alone. It is also constrained by the biological conditions encoded in the guard function.

For each transition t , we define the effective firing rate by

$$a_t(M) = g_t(M)\lambda_t(M),$$

where

- $\lambda_t(M)$ is the intrinsic propensity derived from the corresponding ODE term;
- $g_t(M) \in \{0, 1\}$ is the guard function associated with transition t .

Equivalently,

$$a_t(M) = \begin{cases} \lambda_t(M), & \text{if the guard condition is satisfied,} \\ 0, & \text{otherwise.} \end{cases}$$

The quantity $a_t(M)$ is the rate used by Gillespie's algorithm. A transition whose guard evaluates to zero has effective rate zero and therefore cannot be selected.

2.3 Role of Guard Functions

The introduction of guard functions is motivated by the threshold-based regulatory logic of the cXR-tXA model. In the PADE interpretation, the state space is partitioned into regions separated by thresholds. Within each region, a regulatory interaction is considered either active or inactive. These conditions determine whether a biological reaction is admissible from the current state.

One may wonder whether guard functions are necessary, since the Hill functions in the original ODE already decrease production rates when regulatory conditions are unfavorable. However, a small propensity and a zero propensity have fundamentally different stochastic meanings.

As long as

$$\lambda_t(M) > 0,$$

transition t remains possible. Even if its rate is very small, it may eventually fire if the system remains sufficiently long in the same state. Consequently, relying only on continuous Hill-type propensities can allow biologically forbidden transitions to occur with small but non-zero probability.

The guard function enforces a qualitative constraint:

$$g_t(M) = 0 \implies a_t(M) = 0.$$

Thus, the transition is completely removed from the stochastic competition. In this sense, the intrinsic propensity determines *how frequently* an admissible reaction occurs, whereas the guard function determines *whether* the reaction is admissible at all.

This distinction is important for the qualitative behavior of the model. Preliminary simulations without guard functions produced trajectories that did not reproduce the expected dynamics. Although some propensities became very small near regulatory thresholds, they remained strictly positive, allowing reactions that should be excluded by the PADE logic to occur. Introducing guards eliminated these spurious firings and restored the expected threshold-dependent behavior.

2.4 Why Guards Are Not Encoded Only in the Incidence Matrices

Some threshold constraints could be encoded directly in the Petri net structure through the pre-incidence matrix. For example, if a transition requires the complementary place $\bar{c}_1$ to contain at least θ_c tokens, one may impose

$$\text{Pre}(\bar{c}_1, t) = \theta_c.$$

This structural encoding is appropriate for simple conditions involving a single place. However, several regulatory conditions in the cXR-tXA model depend on combinations of molecular populations. For instance, the activation of *Xist* depends on the combined level of the two trans-acting activators, rather than on each activator independently.

Such conditions are not naturally represented by a single pre-arc. Encoding them purely through the Petri net structure would require additional auxiliary places and synchronization transitions, which would make the model more complex and less readable.

Guard functions provide a more modular solution. They allow threshold conditions to be expressed directly as predicates on the current marking. The incidence matrices remain responsible for the structural evolution of the marking, while the guards encode the biological regulatory logic.

2.5 Stochastic Execution

The use of guard functions does not change the classical stochastic execution principle of the Petri net. At a given marking M , each transition t is assigned an effective rate

$$a_t(M) = g_t(M)\lambda_t(M).$$

Transitions with $a_t(M) = 0$ are excluded from the stochastic competition. The remaining transitions compete according to exponentially distributed waiting times.

If

$$a_0(M) = \sum_t a_t(M)$$

is the total effective propensity, then the waiting time before the next firing is distributed as

$$\tau \sim \text{Exp}(a_0(M)).$$

The probability that the next firing corresponds to transition t is

$$\mathbb{P}(\text{next transition} = t) = \frac{a_t(M)}{a_0(M)}.$$

After the selected transition fires, the marking is updated according to the pre- and post-incidence matrices, and the procedure is repeated from the new marking.

2.6 Summary of the Stochastic Semantics

The proposed stochastic semantics separate three distinct components of the model:

1. the *Petri net structure*, which defines the possible state changes;
2. the *intrinsic propensities* $\lambda_t(M)$, which encode the reaction kinetics inherited from the ODE model;
3. the *guard functions* $g_t(M)$, which encode threshold-dependent biological admissibility conditions.

The effective firing rate is therefore

$$a_t(M) = g_t(M)\lambda_t(M).$$

This formulation preserves the biological meaning of the original ODE system, enforces the qualitative threshold logic of the PADE model, and provides a well-defined stochastic semantics compatible with Gillespie simulation.

3 Simulation

Once the structure of the Stochastic Petri Net and its stochastic semantics have been defined, the dynamics of the system are generated using Gillespie's Stochastic Simulation Algorithm (SSA). Starting from an initial marking, the algorithm repeatedly identifies the admissible transitions, computes their effective firing rates, samples the time until the next reaction, selects the transition that fires, and updates the marking accordingly.

The simulation proceeds until a prescribed final simulation time is reached or until no transition remains fireable.

3.1 Enabled Transitions

Let M denote the current marking of the Petri net. A transition is considered *enabled* if it satisfies both the structural constraints of the Petri net and the biological constraints imposed by its guard function.

More precisely, a transition t is enabled if

1. all input places contain the required number of tokens, i.e.,

$$M(p) \geq \text{Pre}(p, t) \quad \forall p \in P,$$

where P denotes the set of places;

2. its guard function evaluates to one,

$$g_t(M) = 1.$$

The set of enabled transitions at marking M is denoted by

$$\mathcal{E}(M).$$

For every enabled transition, the effective firing rate is

$$a_t(M) = g_t(M)\lambda_t(M).$$

The total firing rate of the system is therefore

$$a_0(M) = \sum_{t \in \mathcal{E}(M)} a_t(M).$$

If

$$a_0(M) = 0,$$

no reaction is possible and the simulation terminates.

3.2 One Gillespie Iteration

At each iteration, Gillespie's algorithm performs four successive operations.

First, the set of enabled transitions is determined and their effective propensities are evaluated.

Second, the waiting time until the next reaction is sampled according to the exponential distribution

$$\tau \sim \text{Exp}(a_0(M)).$$

Using the inverse transform method,

$$\tau = -\frac{\ln(U_1)}{a_0(M)},$$

where

$$U_1 \sim \mathcal{U}(0, 1).$$

Third, a second independent random variable

$$U_2 \sim \mathcal{U}(0, 1)$$

is used to select the transition that fires. Transition t_i is chosen with probability

$$\mathbb{P}(t_i) = \frac{a_i(M)}{a_0(M)},$$

which is implemented by selecting the unique transition satisfying

$$\sum_{j < i} a_j(M) < U_2 a_0(M) \leq \sum_{j \leq i} a_j(M).$$

Finally, the marking is updated according to the incidence matrix

$$C = \text{Post} - \text{Pre},$$

through

$$M \leftarrow M + C(:, t_i),$$

and the simulation time is advanced,

$$t \leftarrow t + \tau.$$

The procedure is then repeated from the new marking.

3.3 Simulation Algorithm

Algorithm 1 summarizes the simulation procedure.

Algorithm 1 Gillespie simulation of the stochastic Petri net

```

1: Initialize the marking  $M$ 
2: Initialize the simulation time  $t \leftarrow 0$ 
3: while  $t < t_{\max}$  do
4:   Determine the enabled transition set  $\mathcal{E}(M)$ 
5:   Compute the effective propensities  $\{a_t(M)\}_{t \in \mathcal{E}(M)}$ 
6:   Compute

$$a_0(M) = \sum_{t \in \mathcal{E}(M)} a_t(M)$$

7:   if  $a_0(M) = 0$  then
8:     break
9:   end if
10:  Draw  $U_1, U_2 \sim \mathcal{U}(0, 1)$ 
11:  Compute

$$\tau = -\frac{\ln(U_1)}{a_0(M)}$$

12:  Select transition  $t_i$  satisfying

$$\sum_{j < i} a_j(M) < U_2 a_0(M) \leq \sum_{j \leq i} a_j(M)$$

13:  Update the marking

$$M \leftarrow M + C(:, t_i)$$

14:  Update the simulation time

$$t \leftarrow t + \tau$$

15: end while

```

The resulting sequence of markings forms a continuous-time Markov chain (CTMC), whose transition rates are given by the effective propensities of the enabled transitions. Consequently, the stochastic trajectories generated by the simulator constitute realizations of the underlying CTMC associated with the stochastic Petri net.

4 Equivalence with the Standard Semantics of Stochastic Petri Nets

The stochastic execution model presented in the previous sections is not a new simulation scheme, but an implementation of the classical semantics of Stochastic Petri Nets (SPNs). This section establishes its equivalence with the standard race policy and, consequently, with Gillespie's Direct Stochastic Simulation Algorithm (SSA).

In a classical SPN, each enabled transition is equipped with an independent exponentially distributed firing time. The transition associated with the smallest waiting time fires first, after which the marking is updated and the process is repeated. Gillespie's algorithm adopts a different computational strategy: instead of explicitly generating one exponential clock per transition, it samples directly the time of the next reaction and the identity of the transition that fires.

The following result shows that these two procedures induce exactly the same stochastic process.

4.1 Standard SPN Semantics

Let

$$N = (P, T, \text{Pre}, \text{Post}, M_0, \Lambda)$$

be a stochastic Petri net, and let M denote its current marking.

The set of enabled transitions is denoted by

$$\mathcal{E}(M) \subseteq T.$$

For every enabled transition

$$t_i \in \mathcal{E}(M),$$

an independent exponential random variable

$$\tau_i \sim \text{Exp}(a_i(M))$$

is associated with its firing time, where $a_i(M)$ is the effective firing rate defined previously. The next transition to fire is determined by the classical race policy,

$$\tau = \min_{t_i \in \mathcal{E}(M)} \tau_i,$$

and

$$t^* = \arg \min_{t_i \in \mathcal{E}(M)} \tau_i.$$

Consequently, the stochastic evolution of the net is entirely determined by the family of independent exponential clocks associated with the enabled transitions.

4.2 Distribution of the Next Firing Time

We first derive the distribution of the random variable

$$\tau = \min_i \tau_i.$$

Since the exponential clocks are independent,

$$\begin{aligned} \Pr(\tau > t) &= \Pr(\tau_1 > t, \dots, \tau_n > t) \\ &= \prod_{i=1}^n \Pr(\tau_i > t). \end{aligned}$$

Using the survival function of the exponential distribution,

$$\Pr(\tau_i > t) = e^{-a_i(M)t},$$

it follows that

$$\Pr(\tau > t) = \prod_{i=1}^n e^{-a_i(M)t} = e^{-a_0(M)t},$$

where

$$a_0(M) = \sum_{i=1}^n a_i(M).$$

Hence,

$$\tau \sim \text{Exp}(a_0(M)).$$

Therefore, the minimum of the independent exponential clocks has exactly the distribution used in Gillespie's algorithm, where

$$\tau = -\frac{\ln(U_1)}{a_0(M)}, \quad U_1 \sim \mathcal{U}(0, 1).$$

4.3 Distribution of the Winning Transition

We now determine the probability that transition t_k is the first transition to fire.

This event occurs whenever

$$\tau_k < \tau_j, \quad \forall j \neq k.$$

Conditioning on the realization of τ_k yields

$$\Pr(t_k) = \int_0^\infty f_{\tau_k}(t) \Pr(\tau_j > t, \forall j \neq k) dt.$$

Since

$$f_{\tau_k}(t) = a_k(M) e^{-a_k(M)t},$$

and the exponential clocks are independent,

$$\Pr(\tau_j > t, \forall j \neq k) = \prod_{j \neq k} e^{-a_j(M)t} = e^{-(a_0(M) - a_k(M))t},$$

we obtain

$$\begin{aligned} \Pr(t_k) &= \int_0^\infty a_k(M) e^{-a_k(M)t} e^{-(a_0(M) - a_k(M))t} dt \\ &= a_k(M) \int_0^\infty e^{-a_0(M)t} dt \\ &= \frac{a_k(M)}{a_0(M)}. \end{aligned}$$

Thus, the probability that a transition wins the race is proportional to its firing rate, exactly matching the selection rule used by Gillespie's Direct Method.

4.4 Practical Sampling of the Winning Transition

The previous result establishes that, conditioned on the occurrence of the next reaction, transition t_i is selected with probability

$$\Pr(t_i) = \frac{a_i(M)}{a_0(M)},$$

where

$$a_0(M) = \sum_j a_j(M).$$

Rather than explicitly sampling from this discrete probability distribution, Gillespie's Direct Method uses a simple inverse transform procedure.

A random variable

$$U_2 \sim \mathcal{U}(0, 1)$$

is first generated. Multiplying by the total propensity yields

$$R = U_2 a_0(M),$$

which is uniformly distributed over the interval

$$[0, a_0(M)].$$

The interval is then partitioned into consecutive subintervals whose lengths are equal to the effective propensities,

$$[0, a_1(M)),$$

$$[a_1(M), a_1(M) + a_2(M)),$$

$$[a_1(M) + a_2(M), a_1(M) + a_2(M) + a_3(M)),$$

and so on until the total length reaches $a_0(M)$.

Transition t_k is selected whenever

$$\sum_{j < k} a_j(M) < R \leq \sum_{j \leq k} a_j(M).$$

Since R is uniformly distributed over $[0, a_0(M)]$, the probability of selecting transition t_k is simply the ratio between the length of its interval and the total interval length,

$$\Pr(t_k) = \frac{a_k(M)}{a_0(M)},$$

which coincides exactly with the distribution derived analytically in the previous subsection.

Figure 1 illustrates this sampling procedure.

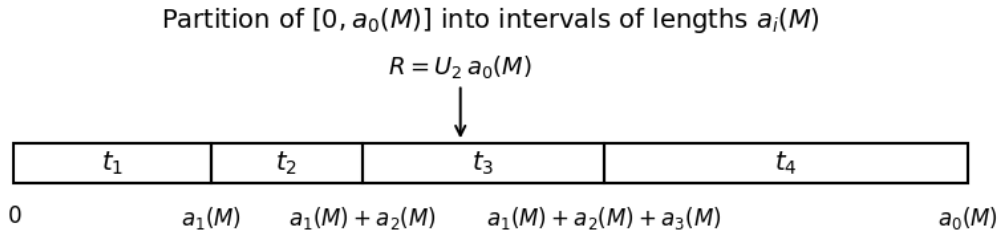


Figure 1: Selection of the next transition in Gillespie's Direct Method. The interval $[0, a_0(M)]$ is partitioned into segments whose lengths are equal to the effective propensities. A uniformly distributed random number determines the transition to fire according to the interval in which it falls.

4.5 Equivalence Theorem

The previous results establish the equivalence between the race semantics of stochastic Petri nets and Gillespie's algorithm.

Theorem 1. *Let M be the current marking of a stochastic Petri net, and assume that every enabled transition t_i is associated with an independent exponential firing time*

$$\tau_i \sim \text{Exp}(a_i(M)).$$

Then the stochastic process generated by the standard SPN race policy is identical in distribution to the process generated by Gillespie's Direct Stochastic Simulation Algorithm using the same effective propensities.

Proof. The previous subsections establish two fundamental properties.

First, the waiting time until the next reaction satisfies

$$\tau \sim \text{Exp}(a_0(M)),$$

where

$$a_0(M) = \sum_i a_i(M).$$

Second, conditioned on the occurrence of the next reaction, transition t_i is selected with probability

$$\Pr(t_i) = \frac{a_i(M)}{a_0(M)}.$$

These are precisely the two random quantities sampled by Gillespie's Direct Method.

Since both approaches generate identical distributions for the waiting time and for the identity of the next transition at every marking, they induce the same probability law over the sequence of visited markings. Consequently, they generate the same continuous-time Markov chain. □

4.6 Continuous-Time Markov Chain Interpretation

The memoryless property of the exponential distribution implies that the future evolution of the system depends only on its current marking. Consequently, the stochastic Petri net induces a continuous-time Markov chain (CTMC).

Each reachable marking corresponds to a state of the CTMC. Whenever a transition t is enabled at marking M , it induces a transition

$$M \longrightarrow M + C(:, t)$$

with rate

$$a_t(M).$$

The infinitesimal generator of the CTMC is therefore completely determined by the effective propensities of the enabled transitions.

Since Gillespie's algorithm produces exactly the same waiting-time and transition-selection distributions as the SPN race policy, it constitutes an exact simulation algorithm for the CTMC associated with the stochastic Petri net.